Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous Institute Affiliated to University of Mumbai)

ESE Examination

Max. Marks: 100 December 2024 Duration: 3Hrs.
Class: T.E. Semester-V Computer Engineering Course Code: CS302
Name of the Course: Software Engineering

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Course Outcomes: At the end of the course student will be able to:
(1) Analyze software requirements.
(2) Apply UML models for a project.
(3) Evaluate system architecture and develop detailed task schedule from the overall estimates and planning.
(4) Illustrate different coding principles with unit test process.
(5) Understand the need for DevOps.

Q. No.		Max Marks	CO-BL
Q.1 A)	Illustrate the phases in the Spiral model of Software Development? Justify why the spiral life cycle model is considered to be a meta model.	10	CO1-2
Q.1 B)	Suppose you are trying to gather the requirements for a software that needs to be developed to automate the activities of supermarket. Identify the main tasks that you would undertake so satisfactorily gather the requirements. Give functional and nonfunctional requirements for the supermarket automation software.	10	CO1-3
Q.2 A)	The software to support a computerized banking network including both human cashiers and automatic teller machines (ATMs) to be shared by a consortium of banks. Each bank provides its own computer to maintain its own accounts and process transactions against them. Cashier stations are owned by individual banks and communicate directly with their own bank's computers. Human cashiers enter account and transaction data. Automatic teller machines communicate with a central computer that clears transactions with the appropriate banks. An automatic teller machine accepts a cash card, interacts with the user, communicates with the central system to carry out the transaction, dispenses cash, and prints receipts. The system requires appropriate recordkeeping and security provisions. The system must handle concurrent accesses to the same	10	CO2-3

	account correctly. The banks will provide their own software for their own computers; you are to design the software for the ATMs and the network. The cost of the shared system will be apportioned to the banks according to the number of customers with cash cards. Write all notations along with their semantics which are used in a Class Diagram. Draw Class diagram for ATM System.		
Q.2 B)	Write all notations along with their semantics which are used in a state machine diagram and Develop UML state machine diagram for arbitrarily deep nesting of parentheses?	10	CO2-3
Q.3 A)	What are the different types of risks that a typical software project might suffer from? Give examples of each type of risk? What are the strategies for risk mitigation?	10	CO3-2
Q.3 B)	Enumerate the different types of cohesion that a module in a design might exhibit. Give examples of each. OR Prepare structured analysis DFD (level 0,1, and level 2) and structured design (structure Chart) for Hostel Management System	10	CO3-3
Q.4 A)	What is Design Pattern? Explain any two Generalised Responsibility Assignment Software patterns (GRASP).	10	CO3-2
Q.4 B)	Write a function for searching an integer value using binary search method. Draw control flow graph. From the control flow graph, determine its Cyclomatic complexity and Find linearly independent paths. Design test cases for the function.	10	CO4-3
Q.5 A)	What do you understand by performance testing? Explain the different types of performance testing.	10	CO4-2
Q.5 B)	What are the four elements that exist when an effective Software Configuration Management system is implemented? Discuss each briefly. OR Discuss DevOps life cycle with example.	10	CO5-2

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